

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA0620	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	B. Shyam raj	Reg. No.:	192421176
Section / Batch:	2024	Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 – 50 Q2 – 50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q1. Explain how algorithms are compared using growth functions. Extend the map by including Input Characteristics and Constraints. Justify why asymptotic analysis is essential for selection. Interpret how this map guides optimal algorithm choice.

Code of Conduct

I, B. Shyam raj(192421176) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Shyam raj

RUBRICS FOR CALCULATION

100%

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts, some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

7-15

12/11/20

Comparing Algorithms Using Growth Functions Asymptotic Analysis of time & space complexity

1. Growth Functions / Asymptotic notation

- Measures how running time or memory use scales as input size n grows, independent of actual runtime.
- Big- O $O(n)$: worst-case upper bound on growth.
- Omega $\Omega(n)$: best-case lower bound on growth.
- Theta $\Theta(n)$: tight bound (average / exact growth).
- Growth rate ranking: $O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n) < O(n!)$.
- Example: merge sort $O(n \log n)$ as n increases, regardless of machine speed.

2. Input Characteristics

- Size (n): number of elements - small datasets vs. very large datasets.
- Structure/data: already sorted, sorted - sorted, random, or nearly - sorted input.
- Distribution: Uniform, skewed, or containing many duplicate keys.
- Data type: numbers, strings, graphs, trees or streaming records.
- Density: Sparse vs. dense data (e.g., sparse graphs/matrices).
- Nature of arrival: static (all at once) vs. dynamic / online (arrives over time).

3. Constraints

- Time constraints: real-time or deadline-bound systems needing predictable speed.
- Space / memory constraints: embedded or memory-limited environments favour in-place algorithms.
- Stability requirement: whether equal elements must keep relative order (e.g., sorting).
- Hardware / platform: availability of parallel or distributed processing.
- Scalability need: System must keep performing well as n grows over time.
- Processing mode: offline (batch) vs. online (incremental) constraints.